**A PROJECT REPORT**

**ON**

# "Ather Copilot"

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**2025-2026**

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This is to certify that the project report entitled **"Ather Copilot"** was successfully completed by the student of sixth semester Diploma in Computer Engineering.

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In partial fulfilment of the requirements for the award of Diploma in Computer Engineering and submitted to the Department of Computer Engineering of S.P.M. Polytechnic Kumathe, Solapur work carried out during the period for the academic year 2025-26 as per curriculum.

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# CHAPTER 9

# OUTPUT

This chapter describes the functional outputs of the Aether Co-Pilot system across its various modules. Due to the interactive nature of the application, the outputs are presented as categorized screenshots of the user interface during active sessions.

## 9.1 THE VISUAL JOURNEY: FROM LANDING TO WORKSPACE

The user experience of Aether Co-Pilot is designed to be a seamless transition from a high-impact landing page to a data-dense productivity workspace.

### 9.1.1 Landing Page and Authentication

The entry point features a "Hero Section" with a dynamic blob animation in the background, signaling the fluid nature of AI.

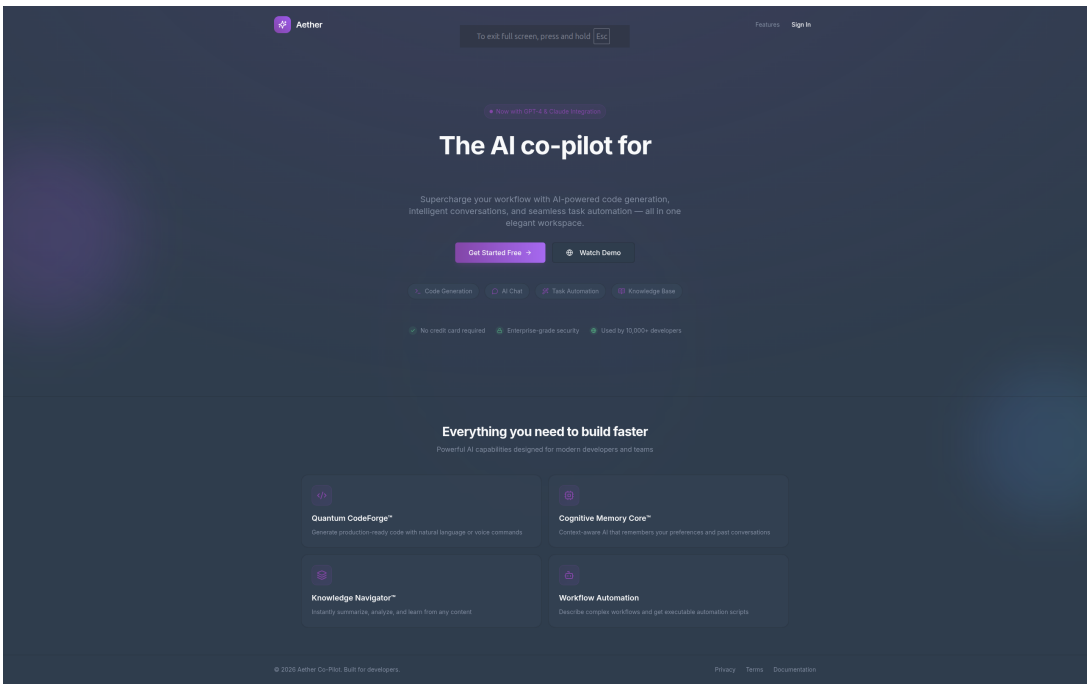


Figure 9.1: The Hero Interface: Features a central CTA with dynamic animations.

### 9.1.2 The Cognitive Dashboard

Upon successful login, the user is greeted by the Dashboard. The productivity grid shows the main feature cards with their respective color-coded gradients. The persistent sidebar illustrates real-time sync indicators and user profile settings.

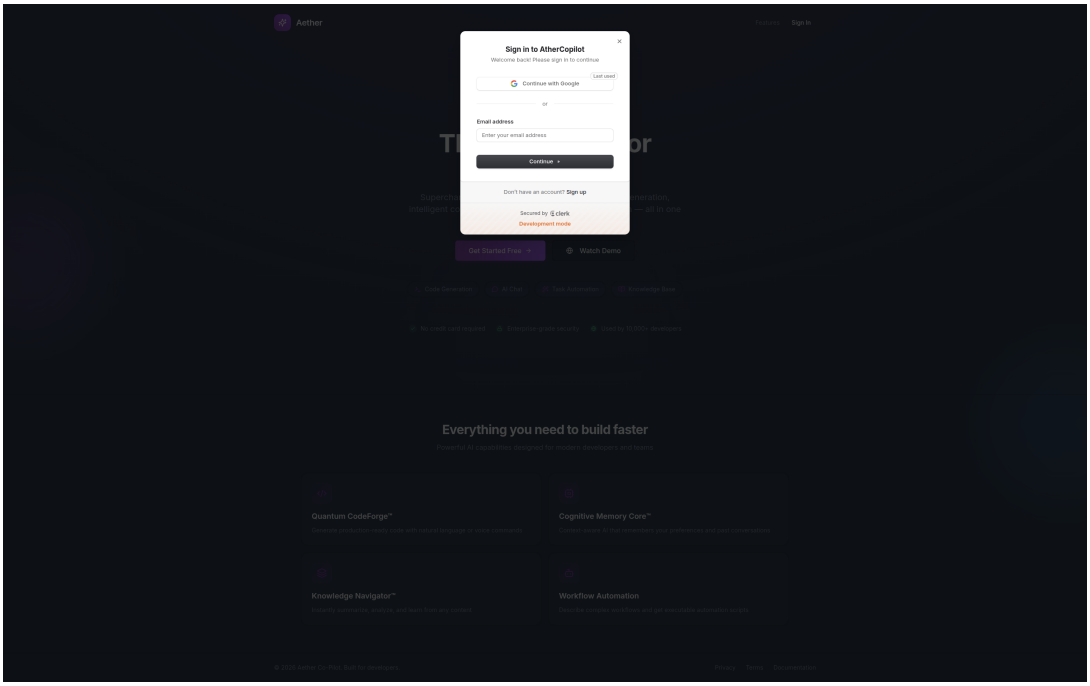


Figure 9.2: Clerk Auth Modal: Displays the custom-branded login interface.

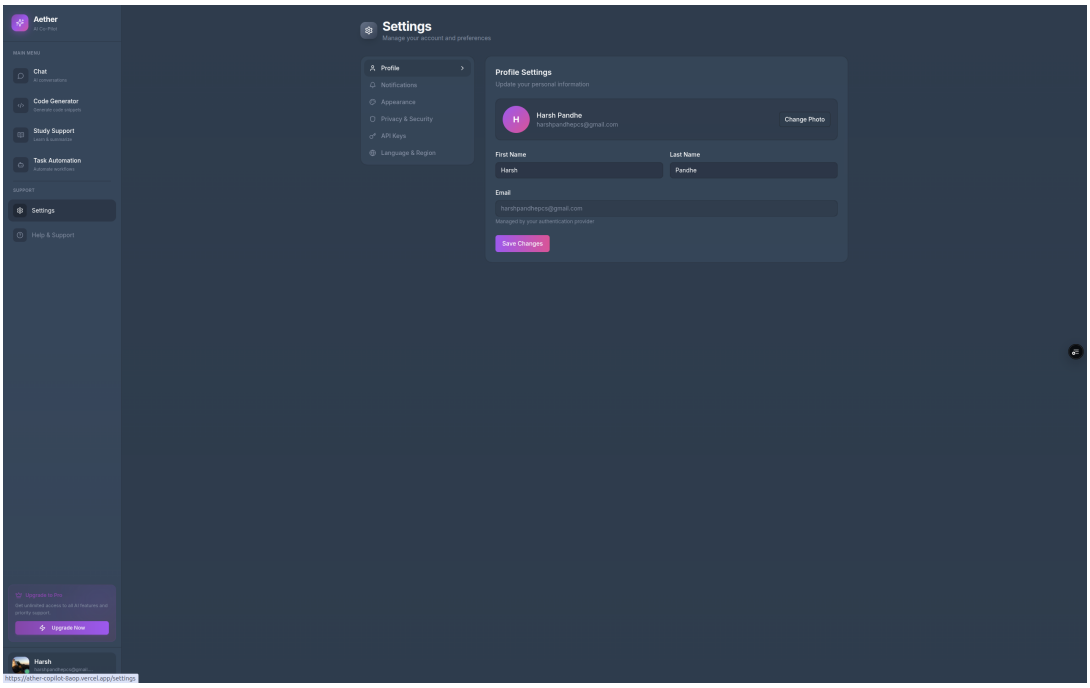


Figure 9.3: User Settings and Activity Sidebar

### 9.1.3 Specialized Tools in Action

## 9.2 USER INTERACTION SEQUENCE: A VISUAL CASE STUDY

To further illustrate the system's output, we documented a complete end-to-step interaction sequence for a "Task Automation" scenario.

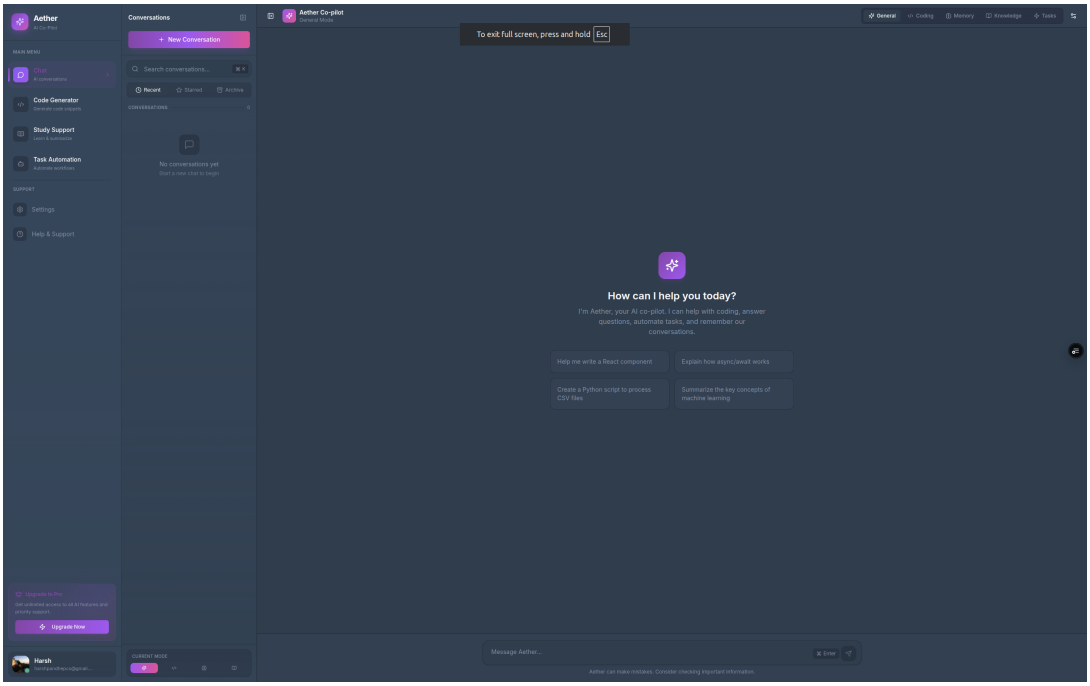


Figure 9.4: Intelligent Chat Interface: Demonstrating message bubbles and the AI interaction mode.

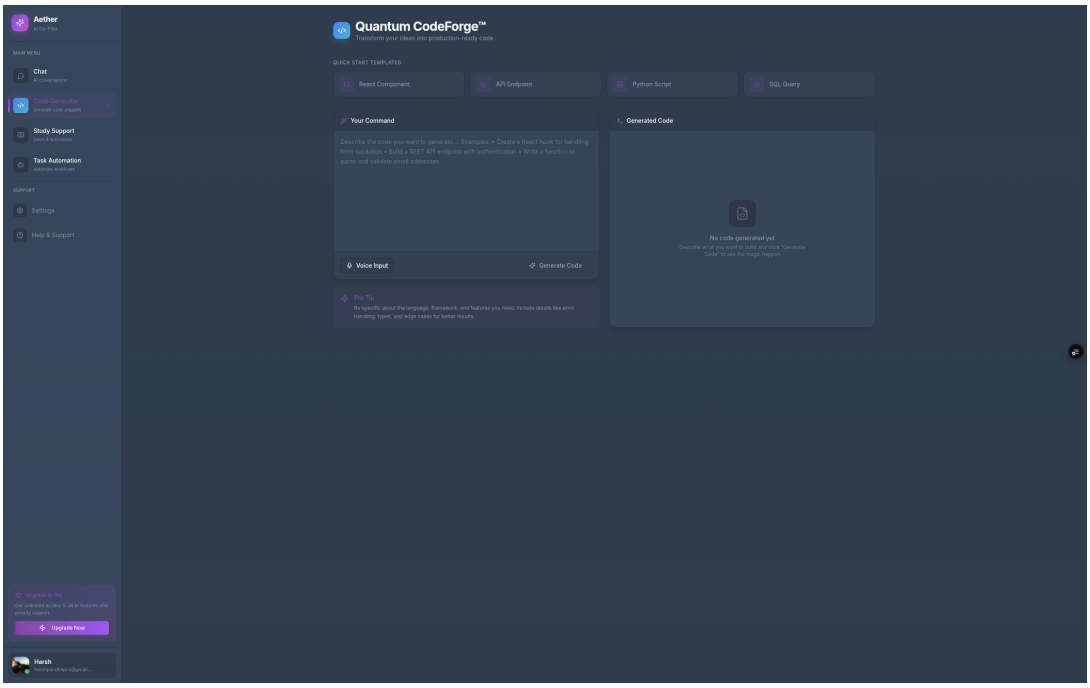


Figure 9.5: Quantum CodeForge: Voice-to-Code generation and live code blocks.

1. **Input Stage**: The user enters the command: "Create a bash script to backup my Desktop to a folder called 'Backup' every Friday at 5 PM."

2. **Processing Stage**: The interface shows a "Genkit Intelligence Thinking..." animation with a pulsing violet glow.

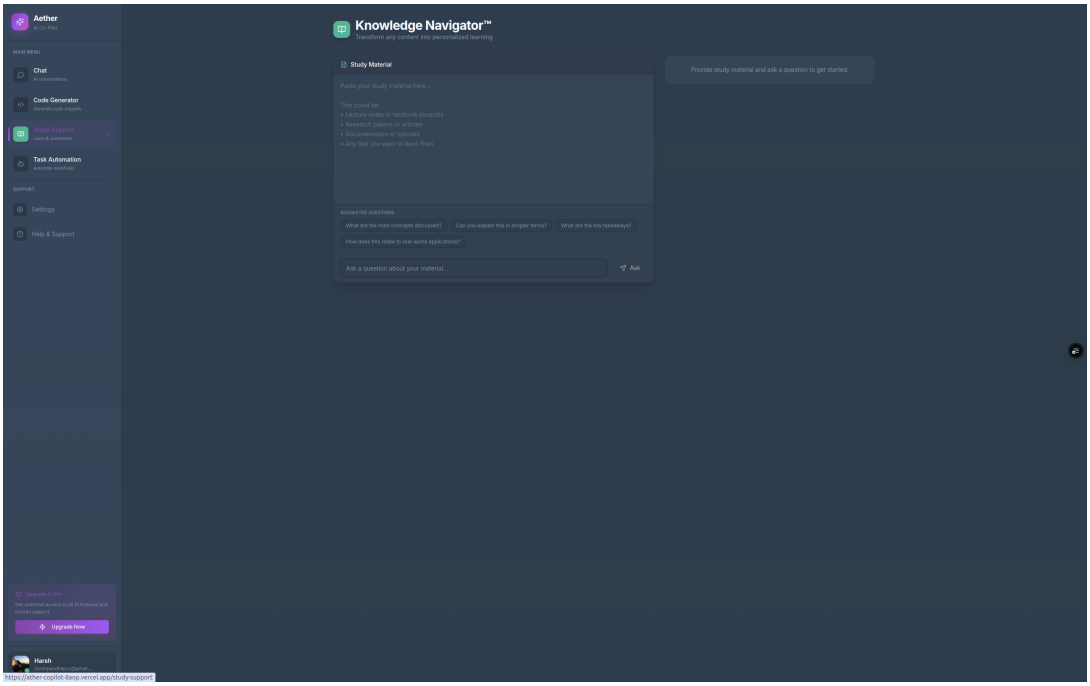


Figure 9.6: Knowledge Navigator: Split-pane layout for PDF study support.

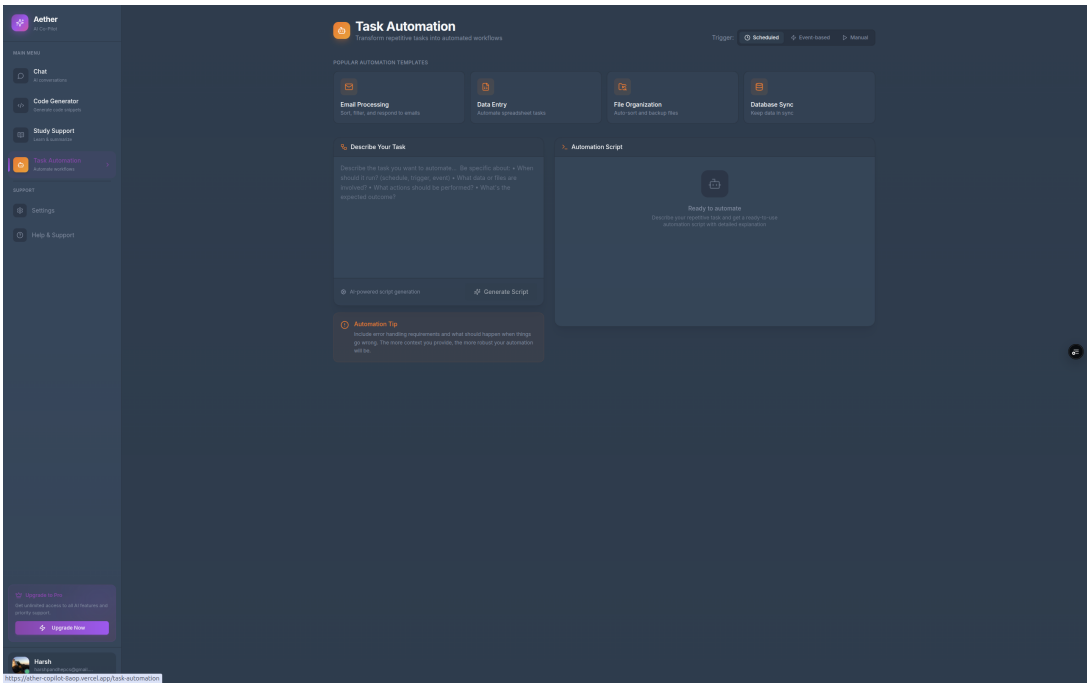


Figure 9.7: Task Automation Engine: Converting natural language to shell scripts.

3. **Visual Output**: The system renders a code block containing the specific ‘crontab‘ entry and a script utilizing ‘rsync‘. A natural language explanation is displayed alongside the code.

4. **Persistence**: The user reloads the navigation sidebar to see the session titled "Backup Script Generation" added to their history with a distinct icon.

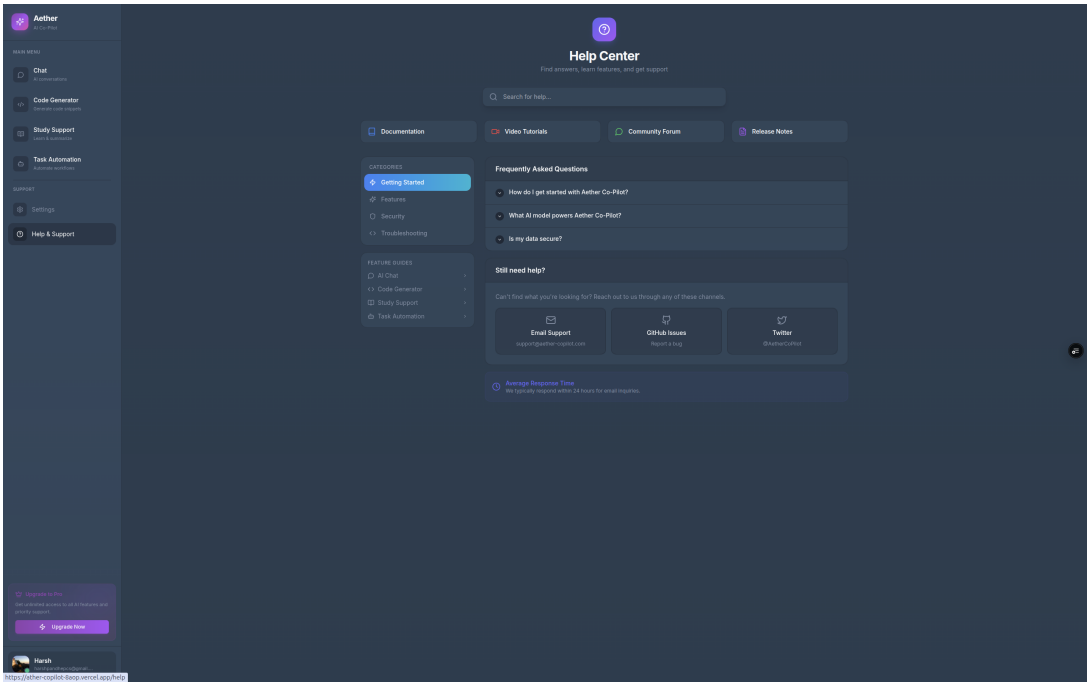


Figure 9.8: Help and Support Interface

## 9.3 ACCESSIBILITY AND INCLUSIVITY IN DESIGN

The visual output of Aether Co-Pilot is designed to be inclusive.

1. **High Contrast Ratios**: The "Aether High-Contrast Dark" theme ensures that text readability meets WCAG 2.1 Level AA standards, particularly for code comments and documentation summaries.

2. **Keyboard Navigation**: All dashboard feature card actions and chat inputs are fully accessible via ‘Tab‘ and ‘Enter‘ keys, catering to users who cannot use a mouse.

3. **Screen Reader Optimization**: Explicit ARIA labels (e.g., ‘aria-label="Send message"‘, ‘aria-label="Record voice command"‘) are used throughout the workspace to ensure that AI-generated artifacts are correctly announced to visually impaired users.

## 9.4 SYSTEM FEEDBACK AND EMOTIONAL DESIGN

We implemented subtle micro-animations using Framer Motion to provide feedback.

- **Loading States**: Skeleton loaders ensure the user is never left with a blank screen during LLM inference.

- **Success Toasts**: Vibrant toast notifications confirm successful data syncs and code generation.

## 9.5 DATA PERSISTENCE OUTPUT

- **Firestore Log**: A screenshot of the Firebase console showing the hierarchical data structure under the ‘/users/‘ collection, validating the security model.